Electromagnetism in Differential Forms

Maxwell Theory

Integral form

We are going to describe the Maxwell equations in integral form. For this, we need a bunch of definitions. Let

- $U \subset \mathbf{R}^3$ be a 3-dimensional region in 3-space
- dv denote the volume element on U
- ∂U denote the (2-dimensional) boundary of U
- $\mathbf{n}$ denote the (outward-directed) unit normal to ∂U
- da denote the area element on ∂U induced from the canonical volume form on $\mathbf{R}^3$ and let $d\mathbf{a} = \mathbf{n} da$.
- $A \subset \mathbf{R}^3$ denote an oriented surface
- $d\mathbf{l}$ denote the line element on ∂A induced from the canonical volume form on $\mathbf{R}^3$

The Maxwell equations are

Coulomb Let $Q(U)$ be the charge contained in U. Then

$$4\pi Q(U) = \int_{\partial U} \mathbf{E} \bullet d\mathbf{a}$$

No monopoles There are no magnetic sources:

$$0 = \int_{\partial U} \mathbf{B} \bullet d\mathbf{a}$$

Ampere Let $\mathbf{j}(\mathbf{A})$ denote the current (charge flux) flowing through the surface A. Then

$$\int_{\partial A} \mathbf{B} \bullet d\mathbf{l} = \frac{4\pi}{c}\mathbf{J}(A) + \frac{1}{c}\frac{\partial}{\partial t}\int_A \mathbf{E} \bullet \mathbf{n}\; da$$

Faraday-Lenz

$$\int_{\partial A} \mathbf{E} \bullet d\mathbf{l} = -\frac{1}{c}\frac{\partial}{\partial t}\int_A \mathbf{B} \bullet \mathbf{n}\; da$$

Traditional Calculus 3 form

Let

- f and g denote functions $\mathbf{R}^3 \to \mathbf{R}$
- $\mathbf{u}$ and $\mathbf{v}$ denote vector fields on $\mathbf{R}^3$
- $(x, y, z) = (x^1, x^2, x^3) = (x^i)_{i=1}^3$ denote Cartesian coordinate functions on $\mathbf{R}^3$

In calculus 3, we learn about the differential vector operators

coordinate basis $\partial_{i\,i=1}^{\,3}$ with $\partial_i = \frac{\partial}{\partial x^i}$. By definition of Cartesian coordinates, the coordinate vectors commute $\partial_i \partial_j = \partial_j \partial_i$.

gradient The gradient $\mathrm{grad}(f)$ or ∇f of a function is a vector field with components

$$\nabla f = (\partial^i f) = (\delta^{ij}\partial_j f) = \begin{pmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \\ \frac{\partial f}{\partial z} \end{pmatrix}$$

divergence The divergence $\mathrm{div}(\mathbf{u})$ or $\nabla \bullet \mathbf{u}$ of a vector field is a scalar defined by

$$\nabla \bullet \mathbf{u} = \frac{\partial u^1}{\partial x} + \frac{\partial u^2}{\partial y} + \frac{\partial u^3}{\partial z} = \partial_i u^i$$

Here and henceforth, repeated indices with one "up" and one "down" are summed (Einstein summation convention).

curl The curl $\mathrm{curl}(\mathbf{u})$ or $\nabla \times \mathbf{u}$ of a vector field is a vector field with components defined by

$$(\nabla \times \mathbf{u})^3 = \frac{\partial u^2}{\partial x^1} - \frac{\partial u^1}{\partial x^2}$$

and cyclic permutations of $^1, ^2, ^3$.

We also learn that these operators are related by integration. In particular, for subsets of $\mathbf{R}^3$ in the notation of the previous section, we have the following theorems:

FTC? Let f be a function on (a neighborhood of) of an oriented curve γ with boundary $\partial\gamma = b - a$, possibly with $b = a$ (no boundary).[1] Then

$$\int_\gamma (\nabla f) = \int_{\partial\gamma} f = f|_a^b = f(b) - f(a)$$

Green Let $\mathbf{u}$ denote a vector field on (a neighborhood of) A. Then

$$\int_A (\nabla \times \mathbf{u}) \bullet d\mathbf{a} = \int_{\partial A} \mathbf{u} \bullet d\mathbf{l}$$

Stokes Let $\mathbf{u}$ denote a vector field on (a neighborhood of) U. Then

$$\int_U (\nabla \bullet \mathbf{u})\ dv = \int_{\partial U} \mathbf{u} \bullet d\mathbf{a}$$

Ever feel like Green's theorem and Stokes' theorem were secretly the same thing?

We now define the charge density ρ and the current density $\mathbf{j}$ so that

$$Q(U) = \int_U \rho dv \qquad \text{and} \qquad \mathbf{J}(U) = \int_U \mathbf{j} dv$$

Exercise 1. *Put the Maxwell equations into differential form:*

Coulomb

$$\nabla \bullet \mathbf{E} = 4\pi\rho$$

No monopoles

$$\nabla \bullet \mathbf{B} = 0$$

Ampère

$$\nabla \times \mathbf{B} = \frac{4\pi}{c}\mathbf{j} + \frac{1}{c}\frac{\partial \mathbf{E}}{\partial t}$$

Faraday-Lenz

$$\nabla \times \mathbf{E} = -\frac{1}{c}\frac{\partial \mathbf{B}}{\partial t}$$

Notice the overwhelming superiority of the differential form notation.

Exercise 2. *Show that if* $\mathbf{B} = \nabla \times \mathbf{A}$ *for some vector field* $\mathbf{A}$, *the it automatically solves the no-monopole condition.*

Show that if we change $\mathbf{A}$ *to* $\mathbf{A}' = \mathbf{A} + \nabla\lambda$ *for any function* λ *then that does not change* $\mathbf{B}$.

The ambiguity in the definition of $\mathbf{A}$ is an example of something called a "gauge ambiguity".

[1] This means that a and b are boundary points of the curve with the curve leaving from a and arriving at b.

Superior mathematical form

The differential "calc 3" notation is clearly superior to the integral form, it is still dirty for various reasons.

Exercise 3. *Restrict the entire universe to two dimensions instead of 3. What does the cross product mean in this 2D universe? Extend to a 4D universe by adding an additional direction to up/down, left/right, forward/backward. Can you define the cross product there?*

The problem is that the existence of the cross product is an accident of 3D. A cross product is an anti-symmetric pairing to two vectors/vector fields to give a third vector (field) and this does not need to exist in arbitrary dimension.

We will return to the reason it exists in 3D eventually, but for now, let's see how we can salvage the Maxwell equations by stripping off the 3D-ness. To do this, we need some formalism.

Definition 1. Let V be a real vector space. A linear form on V is a linear map $V \to \mathbf{R}$. The space of linear forms on V is denoted by $V*$; it is called the dual of V.

Claim 1. *Let $dim_{\mathbf{R}}(V) < \infty$. Then $V*$ is a vector space and $dim_{\mathbf{R}}(V*) = dim_{\mathbf{R}}(V)$.*

Definition 2. Let $\mathbf{u}$ be a vector field on $U \subset \mathbf{R}^3$ and let $x^{i}{}_{i=1}^{3}$ be a set of coordinates on U. In these coordinates

$$\mathbf{u} = u^i(x)\partial_i = u^i(x)\frac{\partial}{\partial x^i}$$

For the functions $u^i : U \to \mathbf{R}$ for $i = 1, 2, 3$ are called the components of $\mathbf{u}$ (in the coordinates x).

Take careful note of the position of the indices in the expressions above: The the component functions of vector fields have their indices "up", whereas the coordinate basis vectors ∂ have their indices "down".

Now suppose we have the component functions $u^{i}{}_{i=1}^{3}$ for some vector field $\mathbf{u}$. I'm getting tired of typing that out, so when the meaning is clear, I will abbreviate this as u^i. Anyway, to map from a set like this to a real function, we need to "get rid of the index": If we have 3 other functions $\omega_i(x)$, we can make

$$\omega(\mathbf{u}) = \omega_i(x)u^i(x)$$

which is a function $U \to \mathbf{R}$, as we wanted when we defined linear forms.

Our conclusion is that under the conditions set out in the Claim, the dual space is spanned by elements with coordinate functions ω_i analogous to the u^i of a vector field, but with the indices in the other position. There is a good reason for this, but for the moment you can just take it as a rule.

If you don't want to hit the "I believe" button, consider the following: Let $L : V \to V$ denote a linear operator on a vector space V, and consider its action on a vector $\mathbf{u} \mapsto L(\mathbf{u})$. This operator does not act on $\mathbf{R}$ so if we consider a linear form $\omega : V \to \mathbf{R}$ then it maps $\mathbf{u} \mapsto \omega(\mathbf{u}) \in \mathbf{R}$.

This implies that L must act on ω in a way that ensures that it has no effect on $\omega(\mathbf{u})$ (since it is an element of $\mathbf{R}$ and L has no effect there.

Exercise 4. [2] *By working in components, compare what the matrix of L does to the components of $\mathbf{u}$ and show that that is not the same as what it does to the components of ω. Note that it* is* related, but not the same.*

A linear form $V \to \mathbf{R}$ is also called a 1-form. When we have a coordinate basis for V denoted by $\partial_i = \partial/\partial x^i$, then there is a natural basis for $V*$ denoted dx^i. (Note the position of the index.) Since 1-forms map vectors to numbers, it must be true that the dxs map the ∂s to numbers. The definition/normalization for Cartesian coordinates is this:[3]

$$dx^i(\partial_j) = \delta^i_j = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{otherwise} \end{cases}$$

Exercise 5. *We can combine all theses things into the following synthesis: Let $\mathbf{u}$ be a vector field and let ω be a 1-form field, both defined on $U \subset \mathbf{R}^n$ for general n. Let x^i denote coordinates on U. In these coordinates, $\mathbf{u} = u^i(x)\partial_i$ and $\omega = \omega_i(x)dx^i$ where the repeated indices are summed (as always). Then from these expressions and the definitions show that*

$$\omega(\mathbf{u}) = \omega_i(x)u^j(x)dx^i(\partial_j) = \omega_i(x)u^j(x)\delta^i_j = \omega_i(x)u^i(x)$$

This is an example of a calculation done in a local coordinates called a "component calculation". We will do many of these.

Multilinear algebra

We can make combinations of the basis 1-forms by taking more copies of $V*$ in what is called the "tensor product". Operationally, the tensor product $V*\otimes V*$ is just the collection of objects with basis $dx^i \otimes dx^j$ where the $\otimes$ symbol roughly means "keep them in this order". For example, if $M \in V*\otimes V*$ then what this means is that in local coordinates $M = M_{ij}dx^i \otimes dx^j$. We recognize the components M_{ij} as those of an $n \times n$ matrix.

We can similarly do this for V as well and we can mix V and $V*$. Furthermore, we can make arbitrarily higher numbers of factors like $V \otimes V \otimes V * \otimes V * \otimes V$. Often, we rearrange the factors so that all the vector spaces come first and then all the duals,[4] so we could write this as $V \otimes V \otimes V \otimes V * \otimes V*$.

Exercise 6. *Show that*

[2] *You will probably not know what to make of this question and how to even start, but that's ok. If you think hard, then when we talk about it, it will make more sense and it will be a good investment.*

[3] The symbol on the RHS is called the "Kronecker delta" symbol/tensor.

[4] It is important to note that we are keeping the order in each type of factor, but we are listing all the duals (in their original order) "at the back".

1. *linear operators are elements of* $V \otimes V*$
2. *inner products are elements of* $V * \otimes V*$
3. *the 3D cross product is an element* $V \otimes V * \otimes V*$

Hint: What do you need to provide to each of these to get a scalar?

An element of $V \otimes \cdots \otimes V \otimes V * \otimes \cdots \otimes V*$ with q factors of V and p factors of $V*$ is called a *tensor* of rank (q,p). Tensors of low rank often have special names: A tensor of rank

- $(0,0)$ is a scalar
- $(1,0)$ is a vector
- $(0,1)$ is a linear form or 1-form
- $(0,2)$ is a bilinear form 2-form
- $(0,q)$ is a multi-linear form of rank q or q-form
- $(1,1)$ is a linear operator
- (q,p) with $p+q=2$ are also called matrices

Permutations have a natural action on tensors, and we use them to reduce the general tensors to simpler elements. For example a $(0,2)$ tensor T can be split into symmetric S and anti-symmetric A parts: $T_{ij} = S_{ij} + A_{ij}$. This is so useful that we use a special notation on the indices:

$$S_{ij} = T_{(ij)} = \frac{1}{2}(T_{ij} + T_{ji}) \qquad \text{and} \qquad A_{ij} = T_{[ij]} = \frac{1}{2}(T_{ij} - T_{ji})$$

The anti-symmetric part of $V * \otimes V*$ is so nice that it gets a special name: $V * \wedge V*$. The reason it is nice is related to the following

Exercise 7. *Let* $f : U \to \mathbf{R}$ *be a smooth function and define the Hessian matrix of* f *by* $H(f)$ *as the matrix of second partial derivatives of* f*. Show that* $H(f)$ *is in the kernel of the map* $V * \otimes V* \to V * \wedge V*$*.*

An exterior q-form is an element

$$\omega \in \Lambda^q = \bigwedge_{i=1}^{q} V* = V * \wedge \cdots \wedge V*$$

In local coordinates $x^{i}{}_{i=1}^{n}$ on U, an exterior form of rank p can be written as

$$\omega = \omega_{i_1 \dots i_p}(x) dx^{i_1} \wedge \cdots \wedge dx^{i_p}$$

where the component functions $\omega_{i_1 \dots i_p}(x) = \omega_{[i_1 \dots i_p]}(x)$ are totally anti-symmetric under permutation of indices.

Exercise 8. *Let* $\dim(U) = n$*. Show that*

1. $\dim(\Lambda^p) = \binom{n}{p}$, *in particular* $\Lambda^p = \emptyset$ *if* $p > n$.
2. $\Lambda^p \bigwedge \Lambda^q \cong \Lambda^{p+q}$
3. *for* $\omega \in \Lambda^p$ *and* $\eta \in \Lambda^q$, $\omega \wedge \eta = (-1)^{pq}\eta \wedge \omega$, *in particular* $\omega^2 = 0$ *if* p *is odd.*

Together with the fact that a p-form is still a p-form if you multiply by a real scalar, this exercise shows that the space of exterior forms is an algebra; we call it $\Lambda^\bullet$.

Definition 3. Let $d_i : \Lambda^i \to \Lambda^{i+1}$ for $i = 0, \dots, n$ be a set of maps. Such a set of maps is called a differential on U if $d_{i+1} \circ d_i = 0$ for all i. We define an extension of these maps to forms of any degree by defining $d_i : \Lambda^j \to \Lambda^{j+1}$ to be the 0 map whenever $i \neq j$. Finally, using this extension, we define $d : \Lambda^p(U) \to \Lambda^{p+1}(U)$ for any p by $d = \sum_{i=0}^n d_i$.

Here is an important application/example: For each $i, p = 0, ..., n$, let $\omega \in \Lambda^p(U)$ and define

$$d_i(\omega) = \begin{cases} \partial_k \omega_{j_1 \cdots j_i}(x) dx^k \wedge dx^{j_1} \wedge \cdots \wedge dx^{j_i} & \text{if } i = p \\ 0 & \text{otherwise} \end{cases}$$

Exercise 9.

1. *Show that the collection of maps defines a differential on the algebra of exterior forms.*
2. *Show that, for any* p, $d^2 = 0$ *as a map* $\Lambda^p \to \Lambda^{p+2}$.
3. *Let* $\omega \in \Lambda^p(U)$ *and* $\eta \in \Lambda^q(U)$. *Show that* $d(\omega \wedge \eta) = (d\omega) \wedge \eta + (-1)^p \omega \wedge (d\eta)$.
4. *Suppose* U *is an open ball in* $\mathbf{R}^n$ *and let* $\omega \in \Lambda^p(U)$ *be a* p*-form on* U *such that* $d\omega = 0$. *Show that there is a* $(p-1)$*-form* $\lambda \in \Lambda^{p-1}(U)$ *such that* $\omega = d\lambda$.

This exercise has very interesting consequences for mathematics: An algebra of exterior forms equipped with a differential is called a differential algebra, so the first two conditions are saying that there is such a thing embedded in the structure of calculus. The third part of the exercise is interesting, because you cannot drop the qualifier that U is an open ball.

A form $\omega \in \Lambda^p(U)$ is called "closed" if $d\omega = 0$. A form $\omega \in \Lambda^p(U)$ that can be written as $\omega = d\lambda$ for some $(p-1)$-form λ is called "exact". Note that exact forms are always closed because $d(d\lambda) = 0$ by the very construction of the differential. However, closed forms are not always exact.[5]

[5]For example, if $U \cong S^1$ is a circle and $\omega \in \Lambda^1(S^1)$ is a 1-form, then it is automatically closed because $d\omega$ has degree 2, but that must vanish in a space of dimension 1. (See your previous exercise on the dimension.) Now let $\theta : S^1 \to \mathbf{R}$ be a coordinate on the circle. This coordinate cannot be defined everywhere because, for example, the point usually coordinatized as 0 is the same as the one coordinatized by 2π and $0 \neq 2\pi$, even though the point is the same. What we need in such a case is to cover the circle with two different coordinates: If we remove a point of the circle and call the resulting set U_1 and then we go back to the original circle and remove the antipodal point to define U_2, then $\theta_1 : U_1 \to (0, 2\pi)$ and $\theta_2 : U_2 \to (-\pi, \pi)$ are good coordinates. Together, these coordinates account for every point on S^1 which we can think of as the union $U_1 \bigcup U_2$, but alone, they cannot do the job. Now it is true that on U_1, $\omega = d\theta_1$ and on U_2, $\omega = d\theta_2$.

Often in physics, quantities that we refer to as "vectors" are actually 1-forms. For example, while it is true that the gradient ∇f of a function is a vector, the more natural object is actually the 1-form df. The two are related with the dot product: A dot product $\langle \bullet, \bullet \rangle$ is a map $V \otimes V \to \mathbf{R}$ taking two vectors $(\mathbf{u}, \mathbf{v}) \mapsto \langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u} \bullet \mathbf{v}$. This means that if we fill only the first slot, say, with a vector $\langle \mathbf{u}, \bullet \rangle$ then we get a map $V \to \mathbf{R}$, that is, we get an element of $V*$, or in yet other words, we get a 1-form. You should confirm that

$$df = \langle \nabla f, \cdot \rangle$$

In fact, if you think about how we make a gradient, we take the components of $(df)_i = \partial_i f$ and arrange them as a column vector:

$$((df)_x, (df)_y, (df)_z)) = (\partial f/\partial x, \partial f/\partial y, \partial f/\partial z) \longrightarrow \begin{pmatrix} \partial f/\partial x \\ \partial f/\partial y \\ \partial f/\partial z \end{pmatrix}$$

But that means that we are making a vector out of the components of the 1-form![6]

Exercise 10.

1. *Write out the components of the vector* $\mathbf{B} = \nabla \times \mathbf{A}$.
2. *Suppose B is a 2-form field that is closed $dB = 0$. Then "locally" it is exact, so there is a 1-form A such that $B = dA$. Write out the components of this expression.*
3. *Define the Levi-Civita symbol ε^{ijk} for $i, j, k = 1, 2, 3$ as follows*

$$\varepsilon^{ijk} = \begin{cases} +1 & \textit{if } i = 1, j = 2, k = 3 \textit{ or any even permutation} \\ -1 & \textit{if } i = 2, j = 1, k = 3 \textit{ or any even permutation} \\ 0 & \textit{otherwise} \end{cases}$$

   Use this symbol and to relate the components of the 2-form B to those of the vector $\mathbf{B}$.
4. *Use the Levi-Civita symbol to relate $dB = 0$ to the no-monopole equation.*

We are starting to see the superiority of this this approach to the vector approach, but there is much more...

And it is also true that these expressions agree on the "overlap" $U_1 \bigcap U_2 = (-\pi, 0) \bigcup (\pi, 2\pi)$ because there $\theta_2 = \theta_1 - \pi$ and thus, $d\theta_2 = d\theta_1$ give consistent expressions for ω on $U_1 \bigcap U_2$. But there is no function $f : S^1 \to \mathbf{R}$ defined uniquely on every point of the circle for which df is the same form as ω.

[6] In Euclidean space the difference is minor, because the components of the inner product in a cartesian basis are either 0 or 1 and this allows us to identify $\partial^1 f = \partial_1 f$, *et cetera*. However in special relativity, we add time as a coordinate and the inner product for this part is -1 instead of 1. In that case it really does matter whether we mean x^m or x_m for $m = 0, 1, 2, 3$. In geometry or in general relativity the inner product is a complicated collection of functions. Sometimes, there is not even an inner product, so you cannot define the gradient at all!

Special relativity

In special relativity, we treat the time coordinate t on an equal footing with the spacial coordinates x^i. To do this, we define $x^0 = ct$ so that it has the correct units. Then we extend the index range $x^i \to x^m$ with $m = 0, 1, 2, 3$. Note that $\partial_i \to \partial_m$ with $\partial_0 = \frac{1}{c}\frac{\partial}{\partial t}$. The (relativistic) wave operator is defined by

$$= -\frac{1}{c^2}\frac{\partial^2}{\partial t^2} + \nabla^2 = -\frac{1}{c^2}\frac{\partial^2}{\partial t^2} + \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$

It is called the d'Alembertian. Notice that it looks like $(\partial_m)^2$ except that the sign of the temporal component is reversed! For this reason, we define ∂^m to have the components $\partial^0 = -\frac{1}{c}\frac{\partial}{\partial t}$ and $\partial^i = \delta^{ij}\partial_j$ (as before).

The 4D inner product defined is defined to have components

$$\eta_{mn} = \begin{cases} -1 & \text{if } m = 0 = n \\ +1 & \text{if } m = n \neq 0 \\ 0 & \text{otherwise} \end{cases}$$

It is called the "Minkowski metric". The "inverse" metric η^{mn} is defined to have the same components numerically. This is so that it is, in fact, an inverse: $\eta^{mp}\eta_{pn} = \delta^m_n$ for any m and n (including 0).

Exercise 11.

1. *Show that* $= \eta^{mn}\partial_m\partial_n$.
2. *Extend the Levi-Civita symbol in the natural way (*i.e.* so that in particular $\varepsilon^{0ijk} = \varepsilon^{ijk}$ from before), and use it to interpret the no-monopole symbol as the time component (*i.e.* 0-component) of a 4D equation.*
3. *Calculate the spacial components of the equation you just constructed. Which equation is this? What is B_{0i}?*
4. *Define the Levi-Civita symbol with "the indices down" by "lowering the indices" on the original, that is, $\varepsilon_{mnpq} = \eta_{mm'}\eta_{nn'}\eta_{pp'}\eta_{qq'}\varepsilon^{m'n'p'q'}$. What are the values of its components?*[7]

If all has gone well, you have discovered that the electric and magnetic fields are the space-time and the space-space components of a four-dimensional 2-form. This 2-form is called the electromagnetic field tensor, Maxwell tensor, or Faraday tensor. It is usually denoted F. In local coordinates $F = F_{mn}(x)dx^m \wedge dx^n$.

Exercise 12. *Confirm by explicit calculation that two of the Maxwell equations are equivalent to the statement that the Maxwell tensor is a closed 2-form $dF = 0$.*

[7] *Note that you are actually calculating the determinant of the Minkowski metric thought of as a matrix.*

Exercise 13. *Us the inverse metric to raise the indices on the Maxwell tensor* $F^{mn} = \eta^{mm'}\eta^{nn'}F_{m'n'}$*, and use this to figure out how to write the other two Maxwell equations.*

Note that you need the metric to write the second half of the Maxwell equations, but not to write the first half.

Finally, we introduce the "Hodge" operator with components defined by

$$(*\omega)_{m_1...m_{n-p}} = \frac{1}{p!}\eta_{m_1r_1}...\eta_{m_{n-p}r_{n-p}}\varepsilon^{r_1...r_{n-p}s_1..s_p}\omega_{s_1..s_p}$$

Note that you need a metric to define this operation.

Performing this operation twice brings us from p-forms back to p-forms. In even dimensions, it acts on $\Lambda^{\frac{n}{2}}$ as an involution.

Exercise 14. *Show that* $*^2 = \pm 1$ *on* Λ^p *for* p *odd/even, respectively.*

Exercise 15. *Show that the Maxwell equations are*

$$dF = 0 \qquad \textit{and} \qquad d * F = 4\pi\sigma$$

for a closed* (why?) current-density 3-form σ.*

The first equation is sometimes called the Bianchi identity. The second equation is the equation of motion. (Below we will see why.)

Integration

Differential forms can be integrated. To integrate a $1-form$, we need a 1-dimensional subspace of $\mathbf{R}^n$, that is, a curve. To integrate a $2-form$, we need a surface, and we integrate a 3-form over a volume, *et cetera*. The curve, surface, and volume on which we integrate are called p-chains for $p = 1, 2, 3$. (A 0-chain is an ordered collection of points.) The space of p-chains is denoted by Λ_p. Integration is a pairing of Λ_p with Λ^p: We will not go into the details of chains, but you can imagine that a p-chain can have a boundary that is a $(p-1)$-chain. We write the boundary of a chain γ as $\partial\gamma$.

Exercise 16. *Convince yourself that the boundary of a boundary is empty.*

In fact, it turns out that the space of chains can also be reformulated to have the structure of a differential algebra where the differential is the boundary operator. For this to be valid, we need the condition that $\partial^2 = 0$, but that is precisely the statement that the boundary of a boundary is empty. Note that $\partial : \Lambda_p \to \Lambda_{p-1}$ so the differential moves us down in the degree instead of up.

Now let $\omega \in \Lambda^p(U)$, let $\gamma \in \Lambda_{p+1}(U)$. Then

$$\int_\gamma d\omega = \int_{\partial\gamma} \omega$$

This is called Stokes' theorem for reasons that will be clear after the following

Exercise 17.

1. *Set* $p = 0$ *and show that Stokes' theorem is just the fundamental theorem of line integrals.*
2. *Set* $p = 1$ *and show that Stokes' theorem reduces to Green's theorem.*
3. *Set* $p = 2$ *and show that Stokes' theorem is reduces to what is usually called Stokes' theorem in the traditional form of multivariable calculus.*

We can always integrate over all of $\mathbf{R}^n$ provided that all forms die off fast enough at infinity. This is a physical condition that we often assume because our fields represent physical energy and matter and we want the total amount to not be infinite.

Exercise 18. *Let* $\omega \in \Lambda^p(U)$ *satisfying physical fall-off conditions. Show that*

$$\int \omega \wedge *\omega$$

makes sense as an integral over all $\mathbf{R}^n$ *and write it in multi-variable calculus form (that is, in cartesian coordinates for* $\mathbf{R}^n$*). It may be helpful to know that*

$$dx^{m_1} \wedge \cdots \wedge dx^{m_n} = \varepsilon^{m_1 \cdots m_n} d^n x$$

Finally, we apply all this to the Maxwell theory.

Exercise 19. *Solve the first Maxwell equation as* $F = dA$ *for a four-dimensional 1-form* A*. Define the Maxwell action by*

$$S = -\frac{1}{16\pi} \int F \wedge *F + \int A \wedge \sigma$$

where the integral is over all of spacetime. Vary this action with respect to A*, and use the stationary action principle to derive the equation of motion. Change the gauge* $A \to A' = A + d\lambda$*, and show that the Maxwell action is invariant up to a boundary term "at infinity".*

Note that you need the closure of σ for this. Usually, we also require that all the fields die off at infinity because otherwise we would have problems with finiteness of field energy in the universe.

Exercise 20. *Construct a vector current density* $j = j^m(x)\partial_m$ *from the current 3-form* σ*. What is the equation for the* j *analogous to the closure of* σ*?*

Such current is called conserved becuase

Exercise 21. *Define the charge* $q = \int j^0 d^3x$ *where we integrate over all 3-space. Although this charge can depend in principle on time, show that, in fact, the charge is conserved (in time):* $\frac{dq}{dt} = 0$*. Now redo this exercise directly in terms of the 3-form. How are you defining charge?*